F-1 Engine Characteristics

APOLLO 11

Thrust, pounds		
	Sea level	1,522,000
	Vacuum	1,748,200
Specific impulse, seconds		
	Sea level	265.4
	Vacuum	304.1
Chamber Pressure (psia)		1,125
Engine mixture ratio		2.27
Mission duration, seconds		165
Qualification Life		
	Starts	20
	Duration, seconds	2,250
Weight , pounds		18,616
Production Deliveries		98



CP6_0450_Apollo 11 F1-6.ppt

F-1 Engine Basic Features

APOLLO 11

- Pump-fed, liquid-propellant rocket engine
- Propellants: Liquid oxygen (LOX) and RP-1
- Nozzle area ratio 16:1
- Single unit-single shaft turbopump
  - Two stage impulse turbine drive
  - Dual discharge centrifugal fuel pump
  - Dual Discharge centrifugal LOX pump
- Turbine drive power from gas generator
  - Burning main propellants
- Multiple purpose RP-1 Fuel
  - Turbopump bearing lubrication
  - Hydraulic power for engine valve actuators and vehicle thrust vector control actuators
- No auxiliary starting power required
  - simple tank head start



CP6_0450_Apollo 11 F1-7.ppt